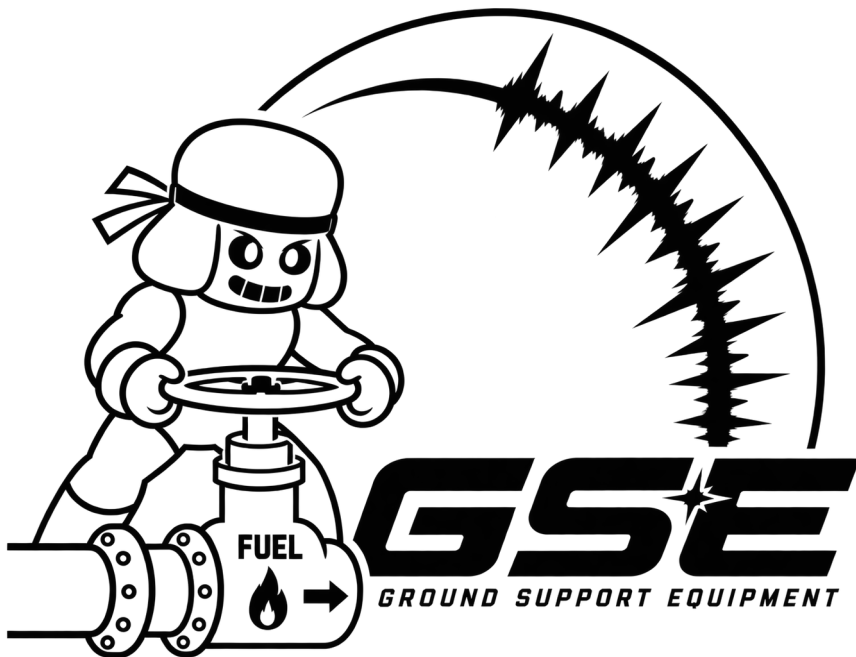


VALVE POWER BOARD

RUBY - REV A



RESPONSIBLE ENGINEER:
NICHOLAS ALVES

REVIEWERS:

SCRT
STUDENT COMPETITION ROCKETRY TEAM

Sheet: /Title Page/
File: top-level.kicad_sch

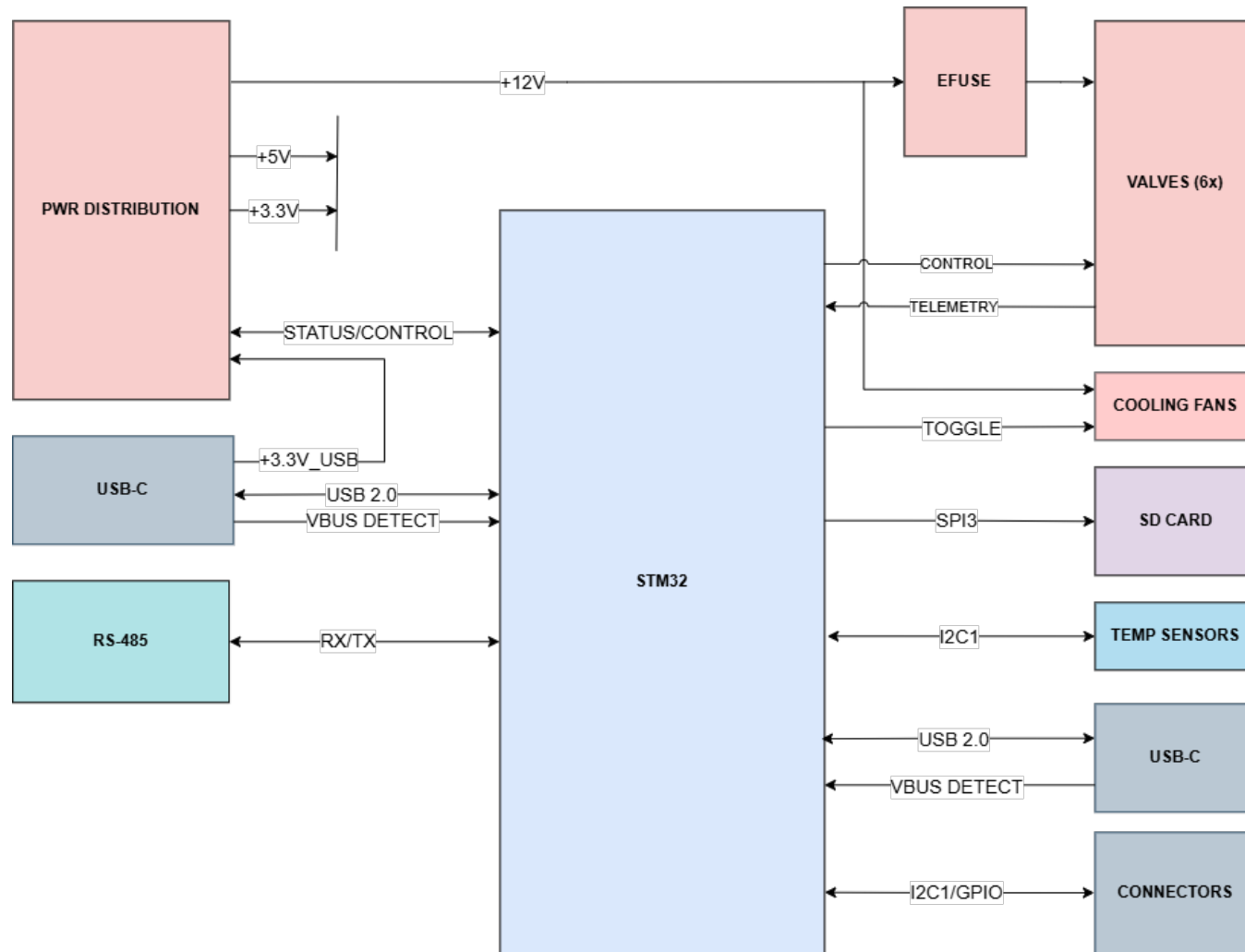
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Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

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Sheet: /Block Diagrams/
File: block_diagrams.kicad_sch

Title: BLOCK DIAGRAM

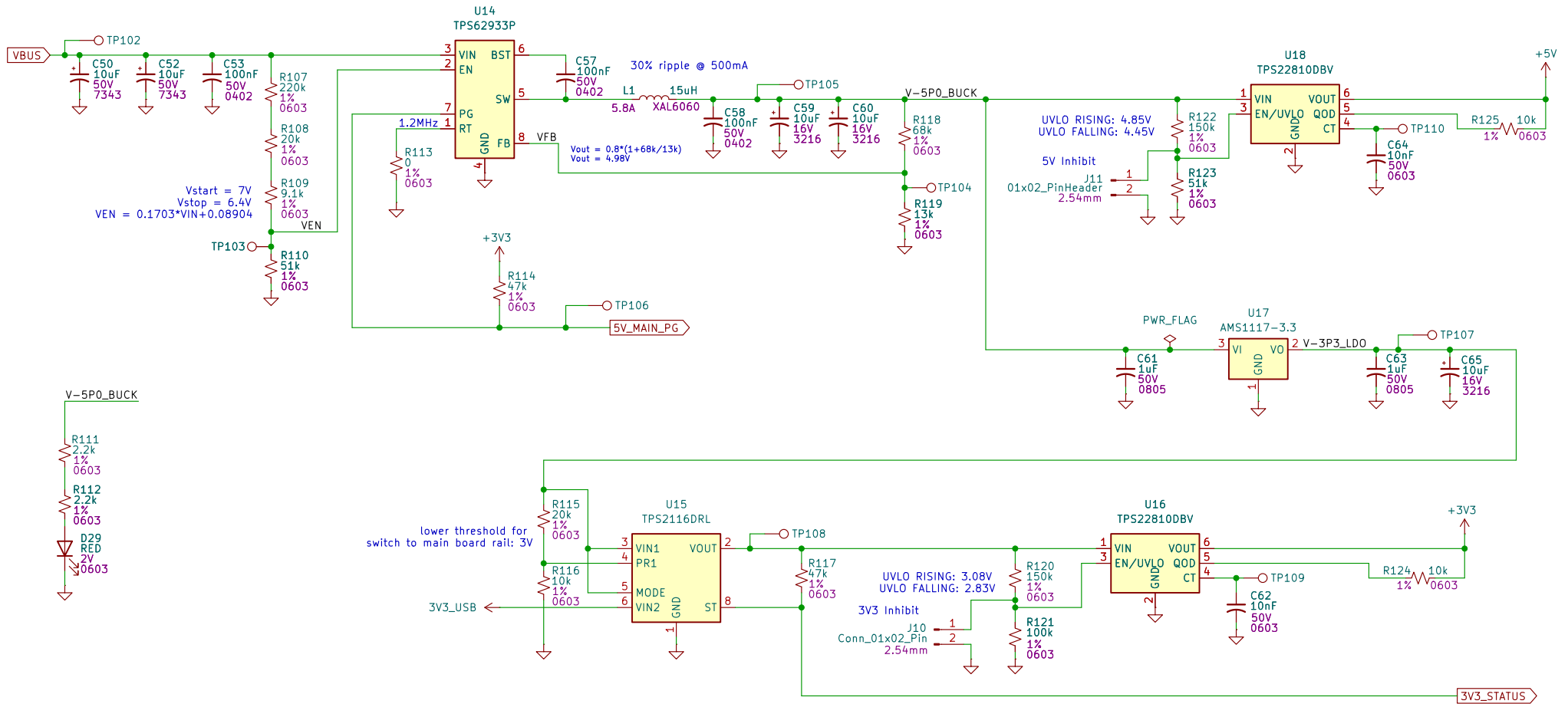
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Rev: A

Id: 2/25

SYSTEM POWER



Sheet: /System Power/
File: system_power.kicad_sch

Title: SYSTEM POWER

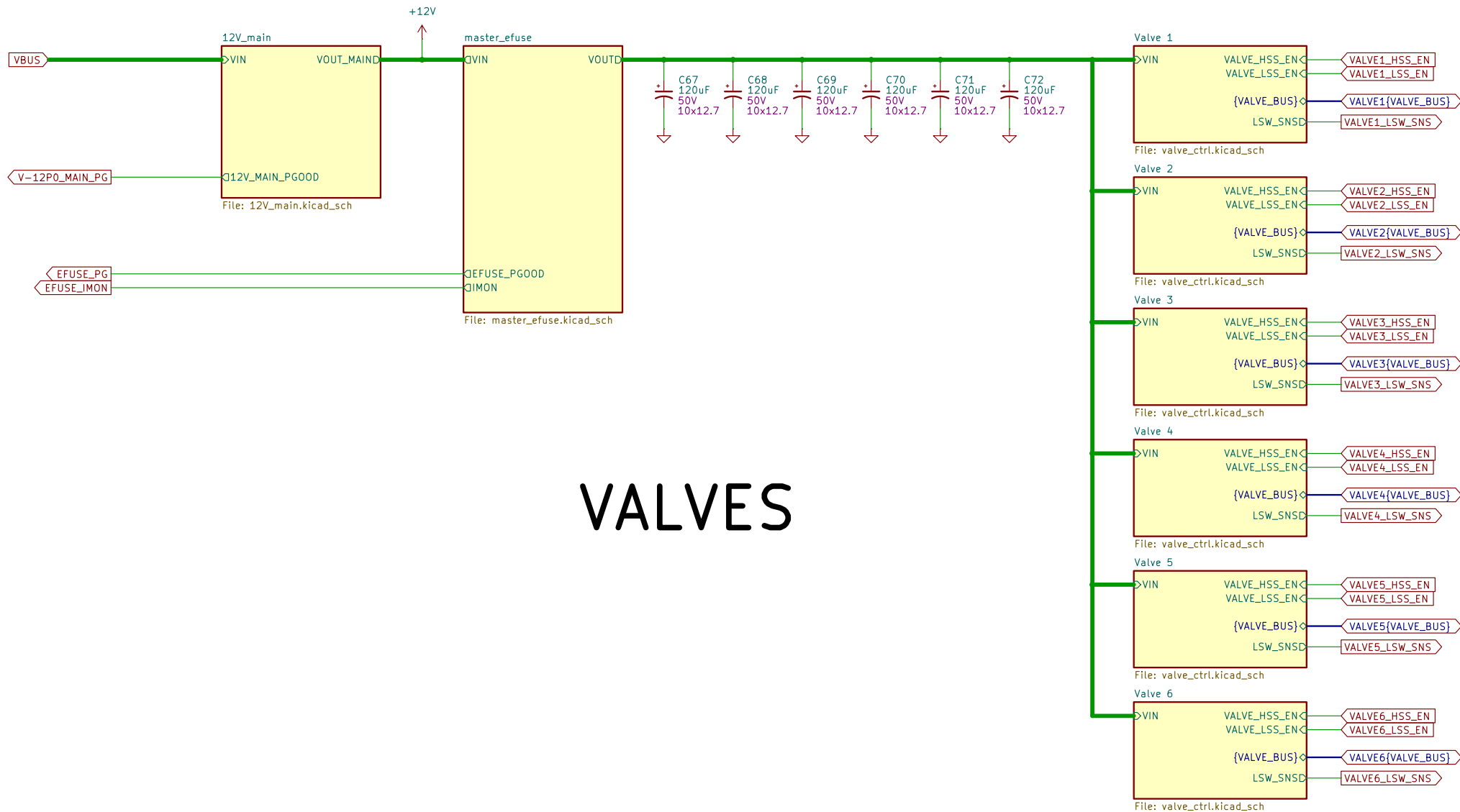
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Id: 3/25

VALVES



Sheet: /Valves/
File: valves.kicad_sch

Title: VALVES

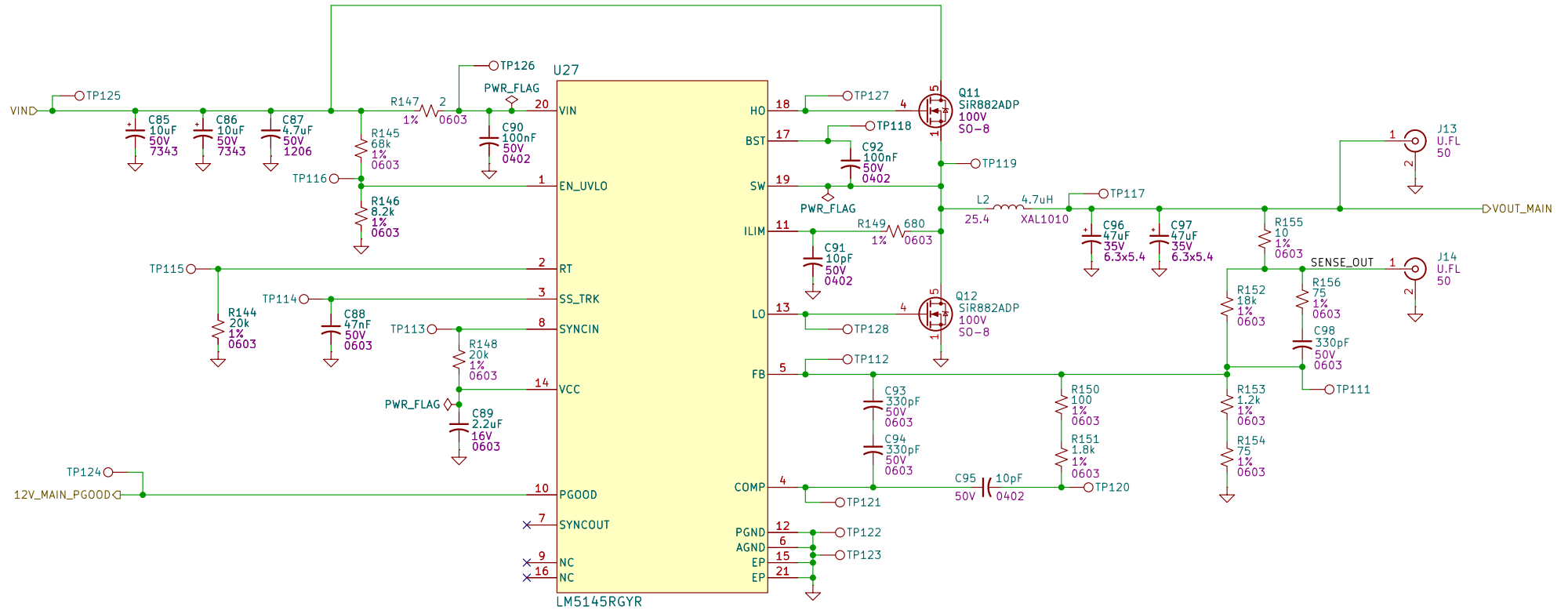
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KiCad E.D.A. 10.0.5

Rev: A

Id: 4/25

12V MAIN



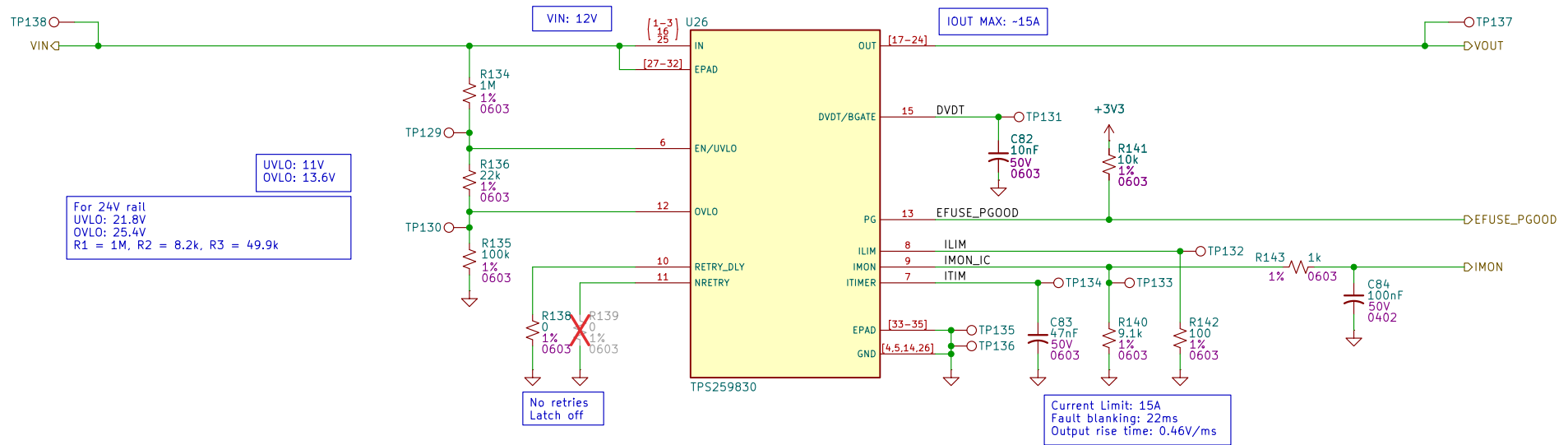
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File: 12V_main.kicad_sch

Title: 12V MAIN

Size: A4 Date: 2026-08-27
KiCad E.D.A. 10.0.5

Rev: A
Id: 5/25

EFUSE



Sheet: /Valves/master_efuse/
File: master_efuse.kicad_sch

Title: EFUSE

Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 6/25

Valve Enable Procedure:

1. Enable common ELOAD
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)

12V, 2A nominal output
Vmax: 36V
Imax: 10A



Title: Valve Output

Title: Valve Output

Size: A4	Date:
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Size: A4	Date: 2026-08-27
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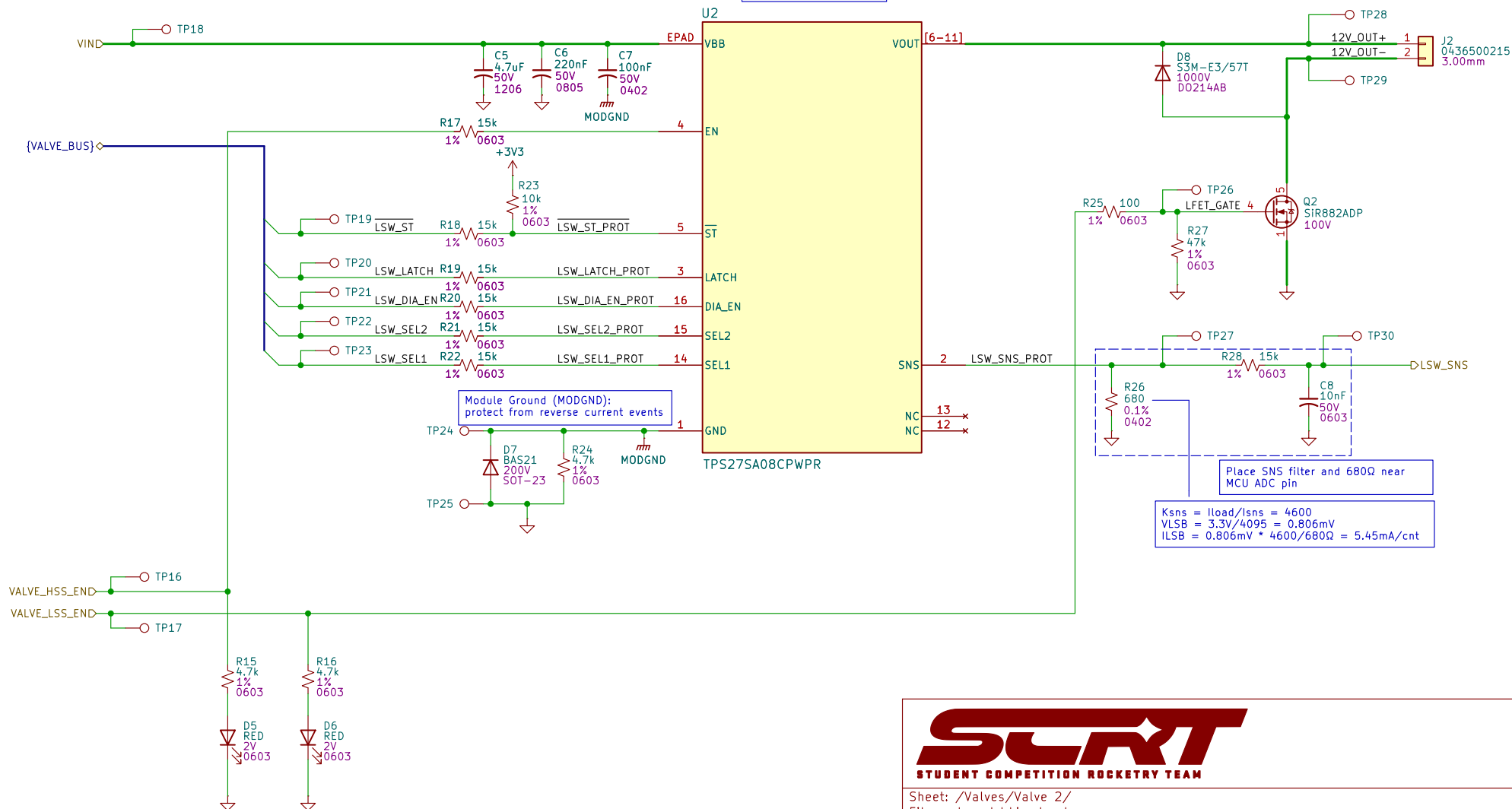
Rev: A

Id: 7/25

Valve Enable Procedure:
 1. Enable common ELOAD
 2. Enable low-side switch (LSS)
 3. Enable high-side switch (HSS)

VALVE OUTPUT POWER

12V, 2A nominal output
 Vmax: 36V
 Imax: 10A



Sheet: /Valves/Valve 2/
 File: valve_ctrl.kicad_sch

Title: Valve Output

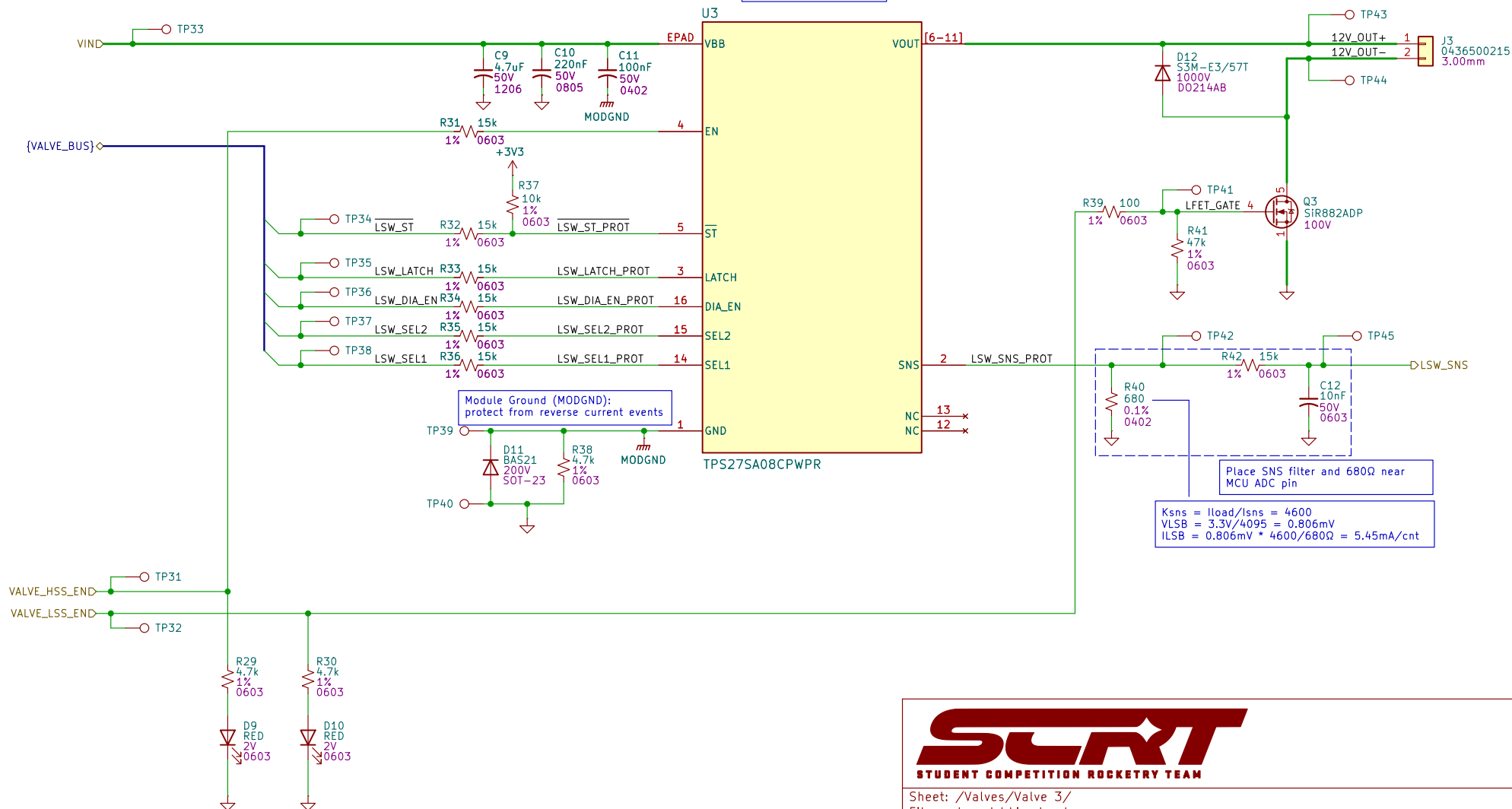
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 KiCad E.D.A. 10.0.5

Rev: A
 Id: 8/25

Valve Enable Procedure:
 1. Enable common ELOAD
 2. Enable low-side switch (LSS)
 3. Enable high-side switch (HSS)

VALVE OUTPUT POWER

12V, 2A nominal output
 Vmax: 36V
 Imax: 10A



Sheet: /Valves/Valve 3/
 File: valve_ctrl.kicad_sch

Title: Valve Output

Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

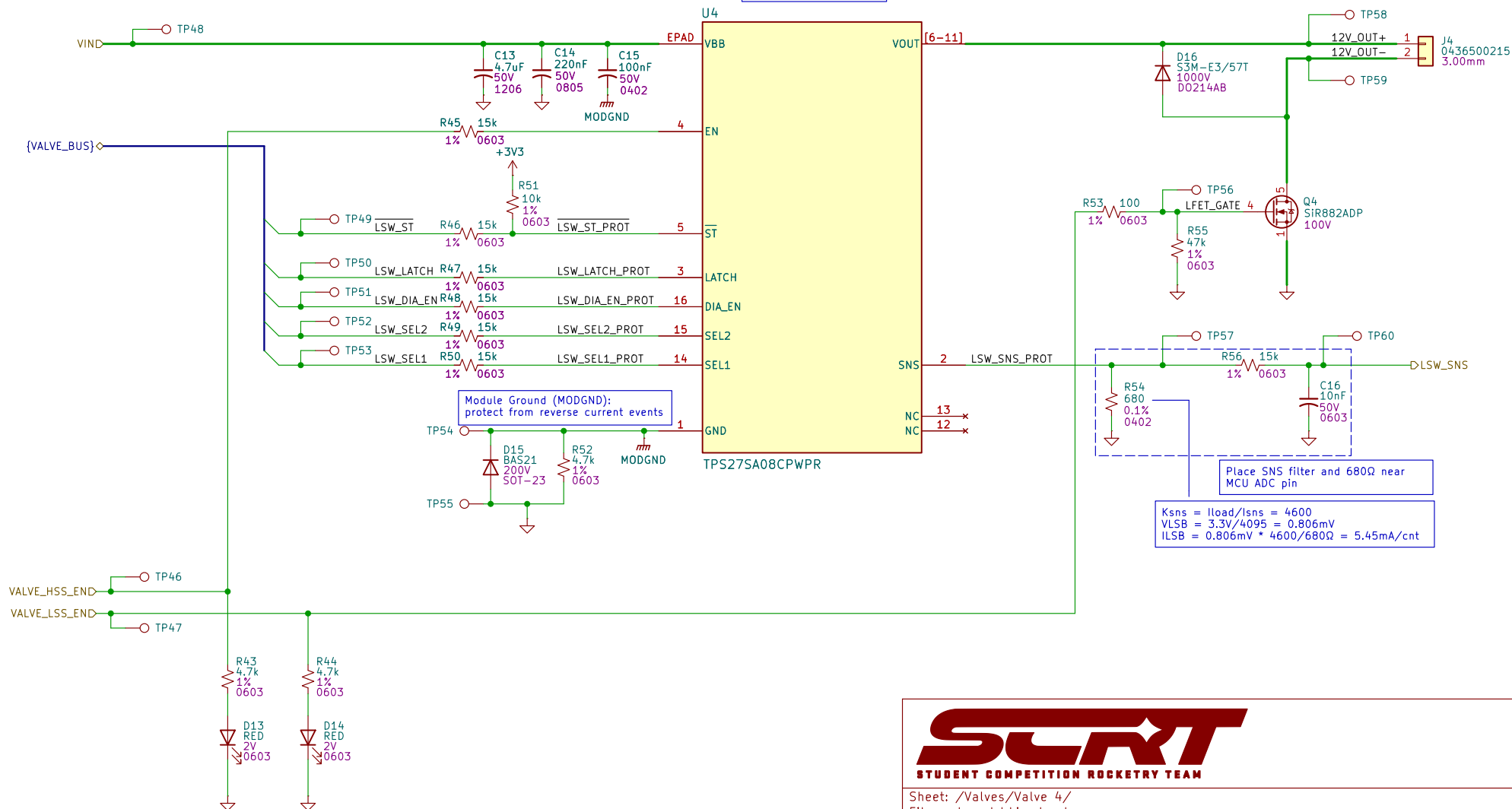
Rev: A

Id: 9/25

Valve Enable Procedure:
 1. Enable common ELOAD
 2. Enable low-side switch (LSS)
 3. Enable high-side switch (HSS)

VALVE OUTPUT POWER

12V, 2A nominal output
 Vmax: 36V
 Imax: 10A



SCRT
 STUDENT COMPETITION ROCKETRY TEAM

Sheet: /Valves/Valve 4/
 File: valve_ctrl.kicad_sch

Title: Valve Output

Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 10/25

Valve Enable Procedure:

1. Enable common ELOAD
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)

12V, 2A nominal output
V_{max}: 36V
I_{max}: 10A



Title: Valve Output

Title: Valve Output

Title: Valve Output

Size: A4	Date:
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Size: A4	Date:
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KiCad E.D.A. 10.0.5

Date: 2026-08-27

Rev: A

Id: 11/25

Valve Enable Procedure:

1. Enable common ELOAD
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)

12V, 2A nominal output
V_{max}: 36V
I_{max}: 10A



Title: Valve Output

Title: Valve Output

Title: Valve Output

Size: A4	Date:
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Size: A4	Date:
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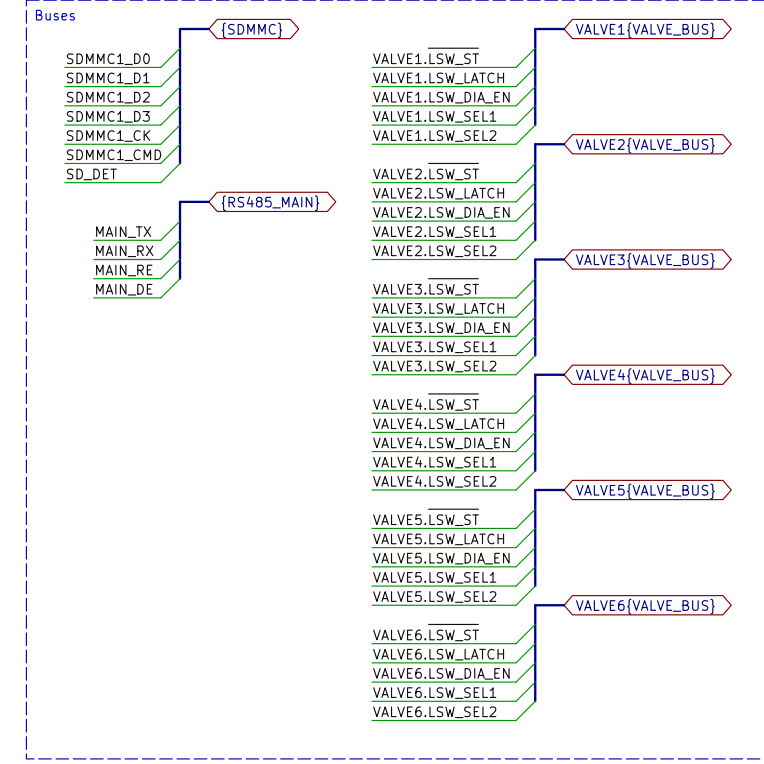
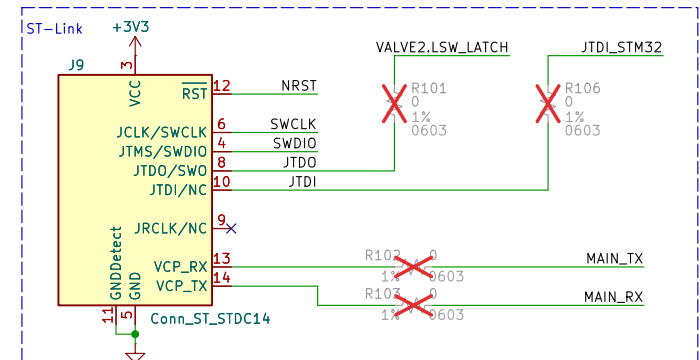
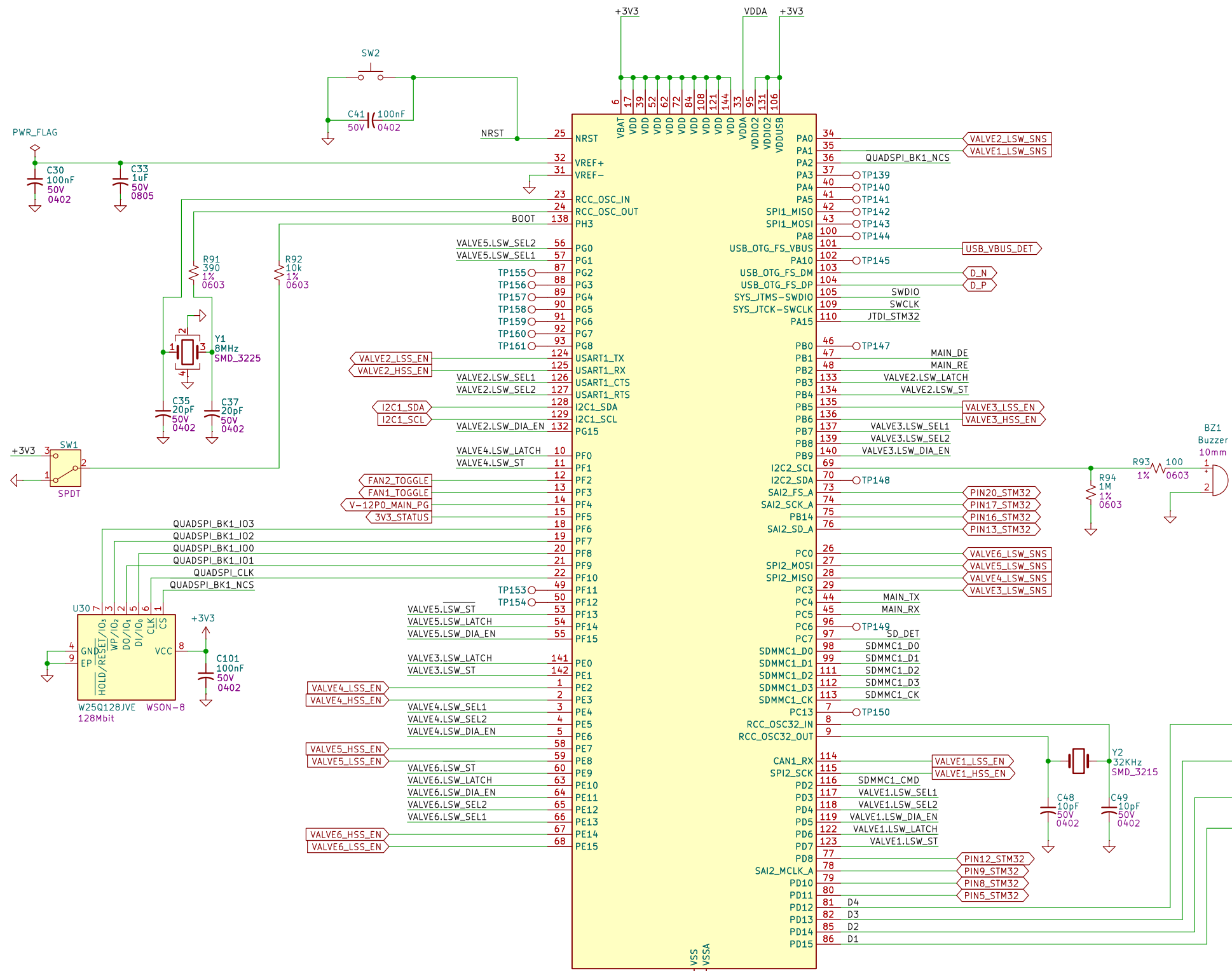
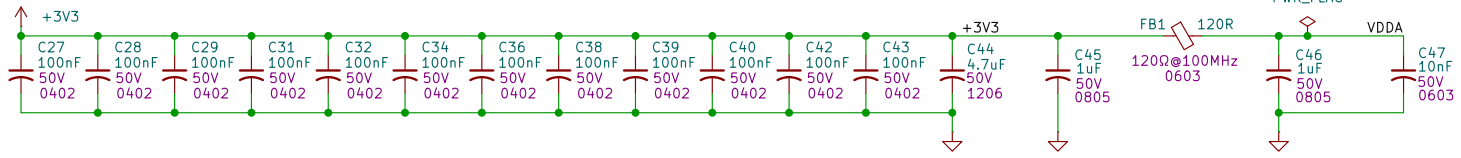
KiCad E.D.A. 10.0.5

Date: 2026-08-27

Rev: A

Id: 12/25

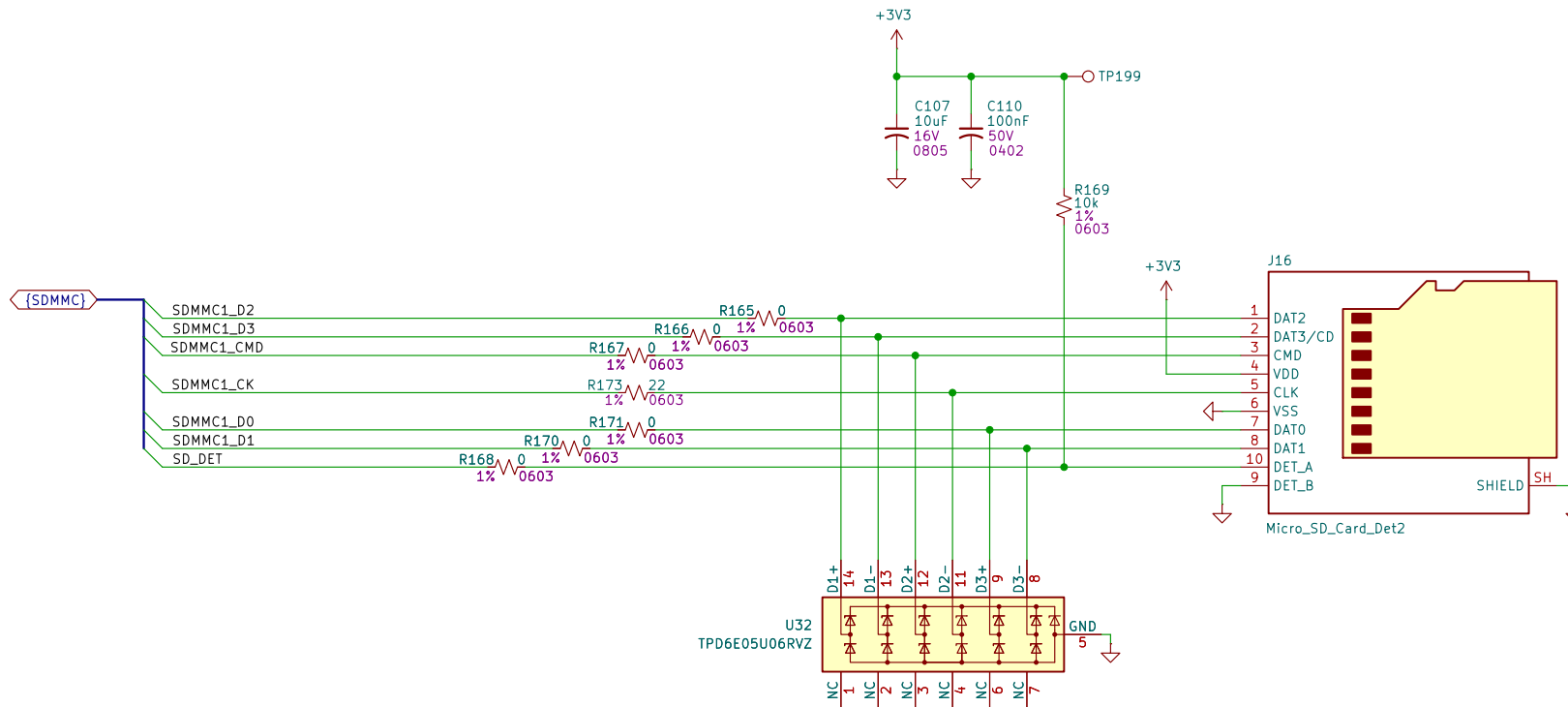
Supply & Analog Supply Filtering



STM32L496ZGT6



SD CARD



Sheet: /SD Card/
File: sd_card.kicad_sch

Title: SD CARD

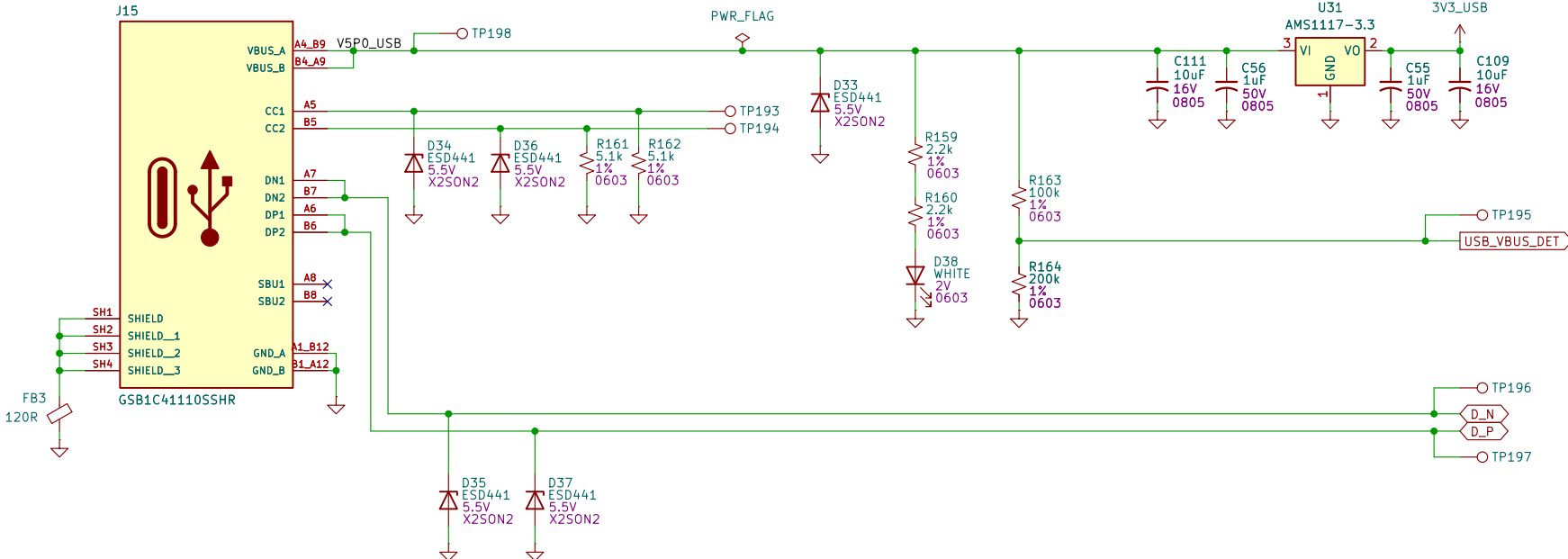
Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 14/25

USB-C



Sheet: /USB-C/
File: usb-c.kicad_sch

Title:

Size: A4

Date: 2026-08-27

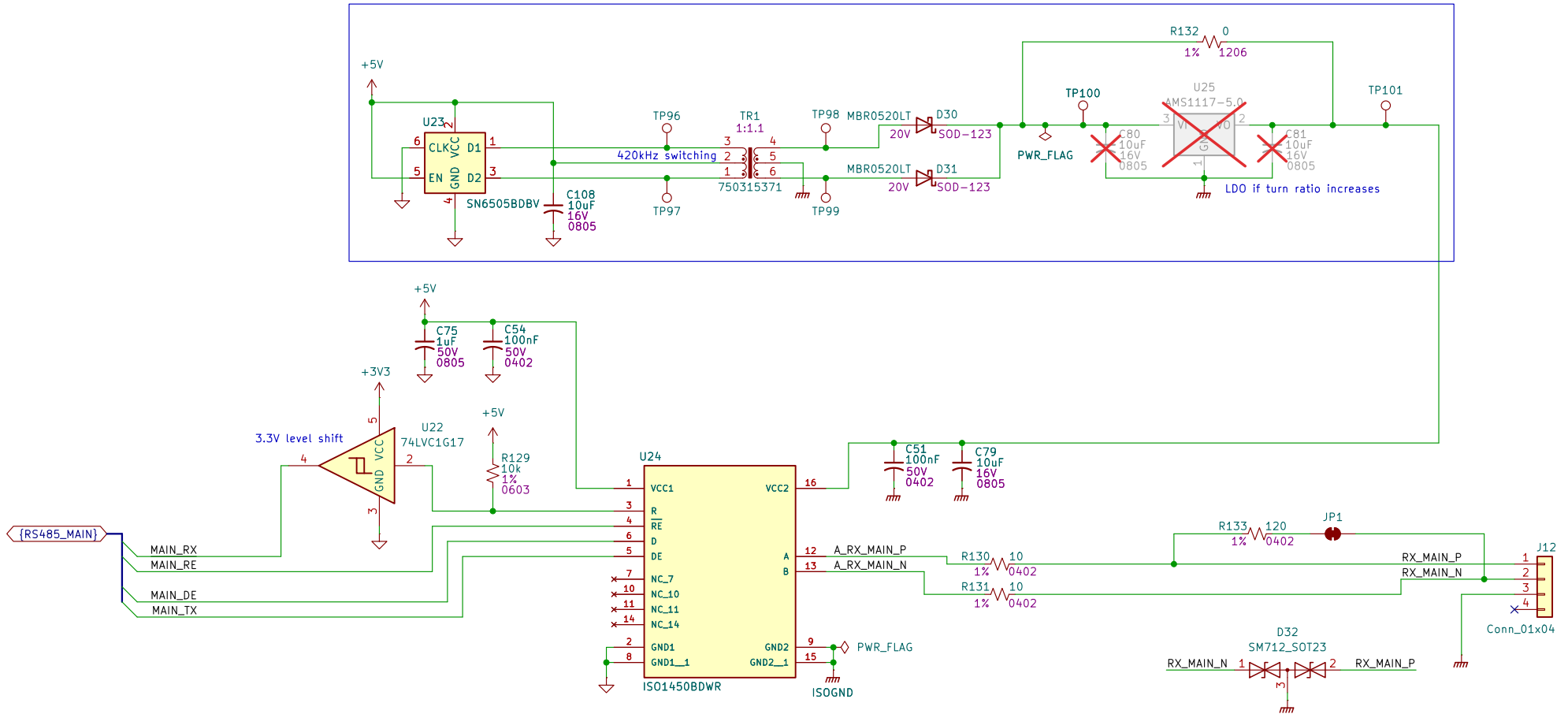
Rev: A

KiCad E.D.A. 10.0.5

Id: 15/25

RS-485

Galvanic Power Isolation



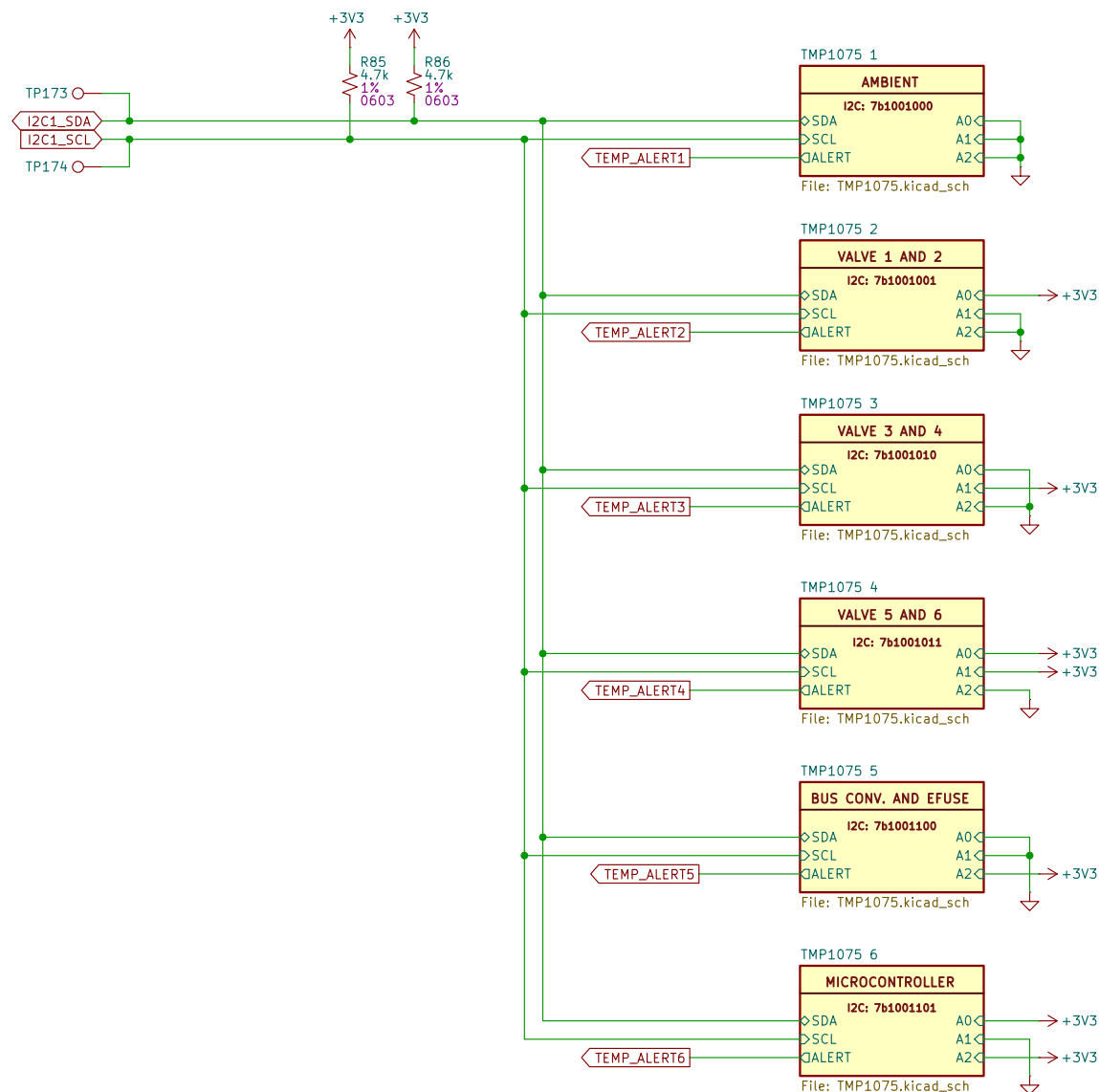
Sheet: /RS-485/
File: rs-485.kicad_sch

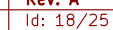
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Size: A4 Date: 2026-08-27
KiCad E.D.A. 10.0.5

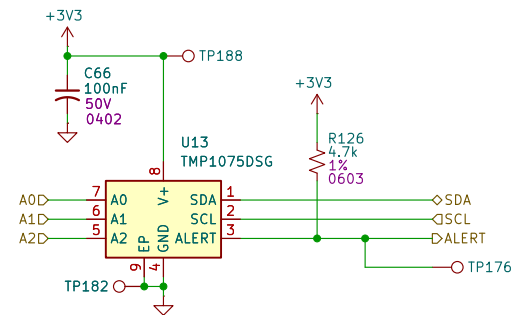
Rev: A
Id: 16/25

TEMPERATURE SENSORS





TMP1075



Sheet: /Sensors/TMP1075 2/
File: TMP1075.kicad_sch

Title: TMP1075

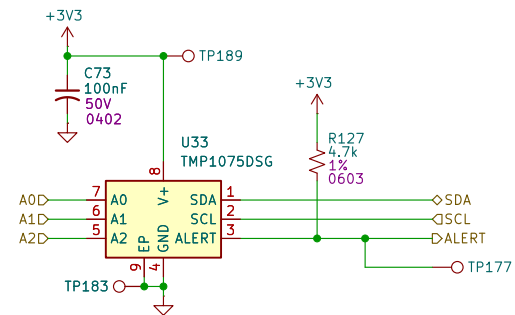
Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 19/25

TMP1075



Sheet: /Sensors/TMP1075 3/
File: TMP1075.kicad_sch

Title: TMP1075

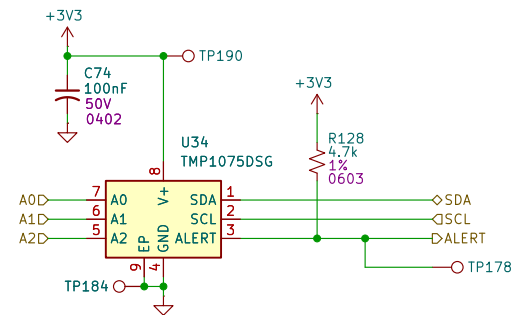
Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 20/25

TMP1075



Sheet: /Sensors/TMP1075 4/
File: TMP1075.kicad_sch

Title: TMP1075

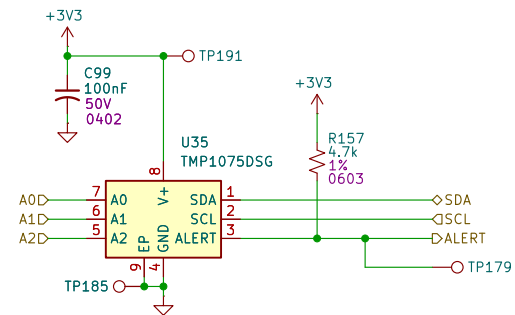
Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 21/25

TMP1075



Sheet: /Sensors/TMP1075 5/
File: TMP1075.kicad_sch

Title: TMP1075

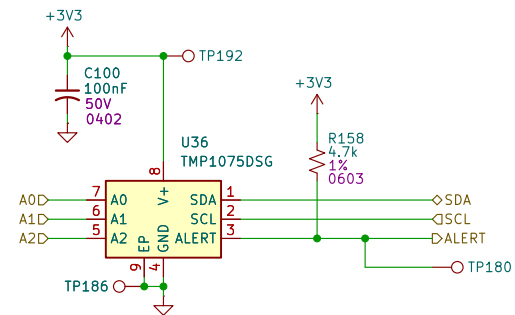
Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 22/25

TMP1075



Sheet: /Sensors/TMP1075 6/
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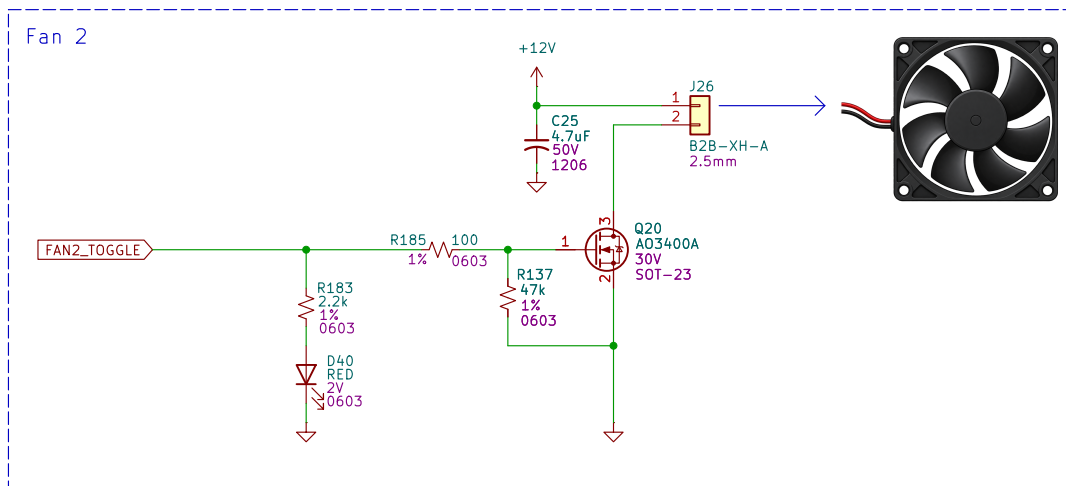
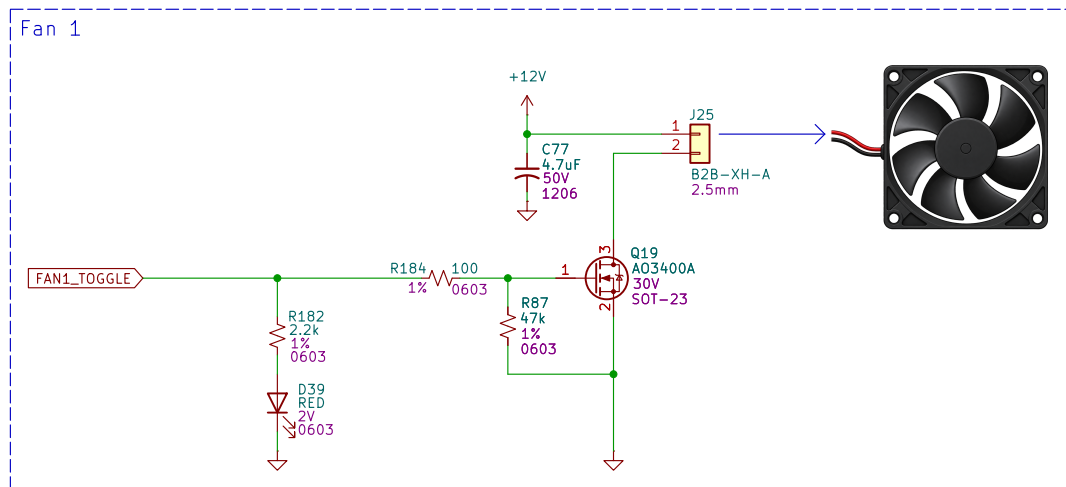
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Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 23/25



Sheet: /Fans/
File: fans.kicad_sch

Title: Fans

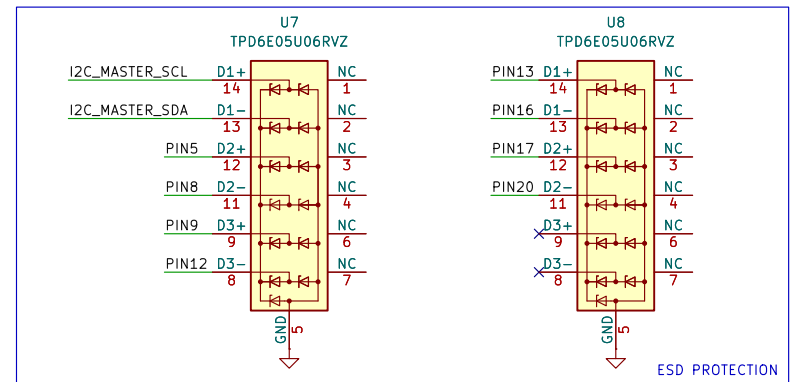
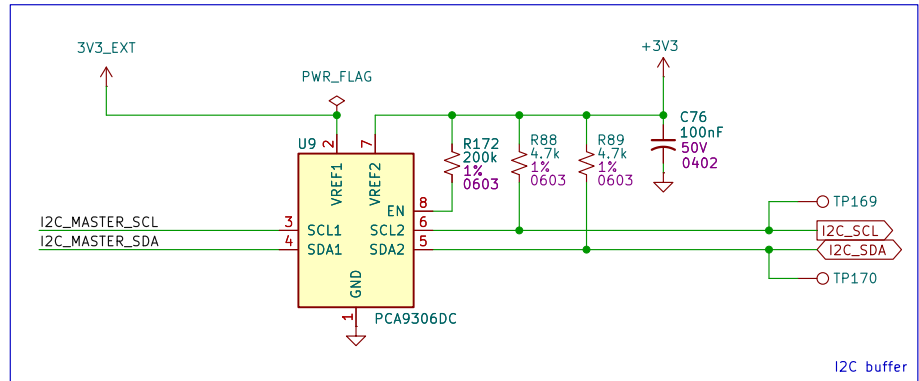
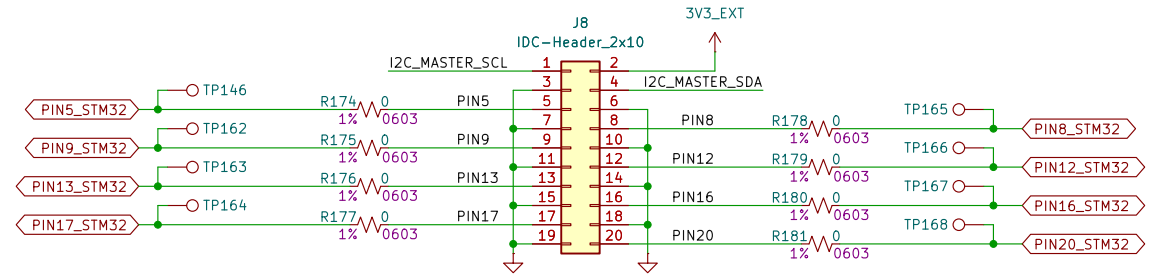
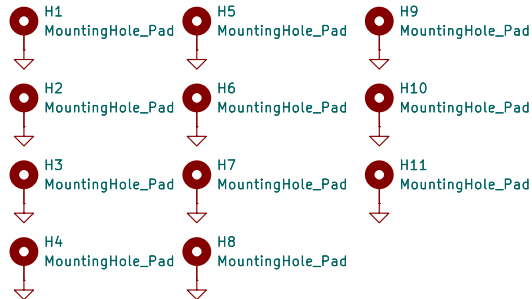
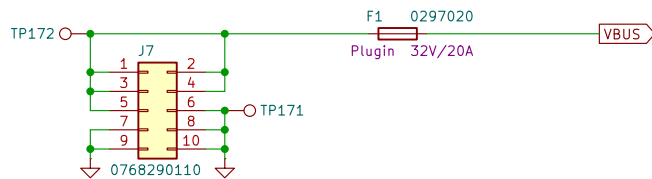
Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: A

Id: 24/25

MECHANICAL



Sheet: /Connectors/
File: connectors.kicad_sch

Title: **MECHANICAL**

Size: A4 Date: 2026-08-27

KiCad E.D.A. 10.0.5

Rev: **A**

Id: 25/25